

PRIVACY-PRESERVING OUTPUTS, ABSTRACT INDICES, AND NON-EXPLICIT DATA REPRESENTATION

Abstract Data Representation Concept

The longitudinal penile physiology monitoring device and associated computational system may present physiologic information through abstracted representations designed to convey meaningful insight without exposing explicit anatomical detail. Data abstraction may transform raw measurements into indices, symbolic outputs, graphical patterns, or non-anatomical visual representations suitable for consumer-facing applications while preserving user privacy.

Abstract outputs may represent physiologic conditions using numerical scores, categorical bands, trend curves, confidence indicators, symbolic icons, geometric representations, or dynamically generated visual signatures derived from multi-modal sensing data.

Privacy-Preserving Visualization

Visualization systems may intentionally avoid explicit anatomical imagery by presenting physiologic interpretation through indirect representations. Such visualizations may include color gradients, abstract waveforms, animated indicators, or algorithmically generated symbols that communicate physiologic state while minimizing sensitive exposure.

Privacy-preserving visualization may support broader adoption in clinical, wellness, consumer, and social environments by allowing physiologic insight to be communicated without revealing personal anatomical characteristics.

Derived Indices and Composite Scores

Derived outputs may include composite indices reflecting rigidity quality, stability behavior, recovery dynamics, readiness state, variability patterns, or longitudinal performance trends. Indices may be computed using sensor fusion algorithms and may incorporate confidence weighting, baseline normalization, and contextual interpretation factors.

Indices may be displayed independently or combined into higher-order profiles representing overall physiologic characterization without exposing underlying raw sensor data.

Symbolic and Profile-Based Representations

In certain embodiments, aggregated physiologic characteristics may be represented through symbolic profiles or dynamically generated visual signatures derived from individualized physiologic patterns. Symbolic representations may evolve over time as longitudinal data is collected and interpreted.

Symbolic or profile-based outputs may be shared across compatible systems, applications, or platforms in anonymized or permission-controlled formats enabling comparison, matching, or compatibility assessment without disclosing raw physiologic measurements.

Interoperability and Data Exchange

Abstract representations may facilitate interoperability with external applications including health platforms, wellness services, or social matching environments through standardized data structures designed to transmit insight rather than sensitive biometric data.

The abstraction and privacy-preserving embodiments described herein are illustrative and non-limiting and encompass present and future techniques for representing wearable-derived physiologic information through non-explicit formats.